LAB ASSIGNMENT – II

1. Display the number of buses from each source.

```
himansh1241 — mysql -u root — 92×13

mysql> SELECT Source, COUNT(*) AS Number_of_Buses
    -> FROM List_Buses
    -> GROUP BY Source;
+------------+-----------------+
| Source     | Number_of_Buses |
+------------+-----------------+
| Pari Chauk |               3 |
| Sector 62  |               1 |
| ISBT       |               4 |
+------------+-----------------+
3 rows in set (0.002 sec)
```

2. Display the number of buses from each source, sorted low to high.

```
himansh1241 — mysql -u root — 92×13

mysql> SELECT Source, COUNT(*) AS Number_of_Buses
    -> FROM List_Buses
    -> GROUP BY Source
    -> ORDER BY Number_of_Buses ASC;
+------------+-----------------+
| Source     | Number_of_Buses |
+------------+-----------------+
| Sector 62  |               1 |
| Pari Chauk |               3 |
| ISBT       |               4 |
+------------+-----------------+
3 rows in set (0.001 sec)
```

3. Display all the details of first 4 cheapest buses from list_buses table

```
himansh1241 — mysql -u root — 92×13

mysql> SELECT *FROM List_Buses
    -> ORDER BY Fare ASC
    -> LIMIT 4;
+--------+-------------+------------+-------------+------+------------+----------+
| Bus_no | Bus_name    | Source     | Destination | Fare | Date       | Time     |
+--------+-------------+------------+-------------+------+------------+----------+
| DL306  | Shatabdi    | ISBT       | Dehradun    | 1007 | 2022-09-29 | 20:45:00 |
| UP301  | Zing bus    | Pari Chauk | Prayagraj   | 1010 | 2022-08-21 | 21:30:00 |
| HR303  | Maheshwaram | ISBT       | Ambala      | 1010 | 2022-09-25 | 07:22:00 |
| HR302  | Cargo       | Sector 62  | Chandigarh  | 1020 | 2022-08-23 | 22:44:00 |
+--------+-------------+------------+-------------+------+------------+----------+
4 rows in set (0.000 sec)
```

4. Display all the passengers whose name begins with D.

```
himansh1241 — mysql -u root — 92×10
mysql> SELECT *FROM Passenger_details
    -> WHERE P_name LIKE 'D%';
+--------------+--------+--------+------+--------+---------+
| Passenger_id | P_name | Bus_no | Age  | Gender | Status  |
+--------------+--------+--------+------+--------+---------+
| SCS012       | Divya  | UP101  |   65 | F      | Waiting |
| SCS014       | Diya   | HR303  |   19 | F      | Confirm |
+--------------+--------+--------+------+--------+---------+
2 rows in set (0.001 sec)
```

5. Write a query that selects the first passenger in alphabetical orderwhose name ends with m.

```
himansh1241 — mysql -u root — 92×11
mysql> SELECT *FROM Passenger_details
    -> WHERE P_name LIKE '%m'
    -> ORDER BY P_name ASC
    -> LIMIT 1;
+--------------+--------+--------+------+--------+---------+
| Passenger_id | P_name | Bus_no | Age  | Gender | Status  |
+--------------+--------+--------+------+--------+---------+
| SCS022       | Alam   | HR302  |   21 | M      | Waiting |
+--------------+--------+--------+------+--------+---------+
1 row in set (0.035 sec)
```

6. Select the average age of male passengers.

```
himansh1241 — mysql -u root — 92×11
mysql> SELECT AVG(Age) AS Average_Age
    -> FROM Passenger_details
    -> WHERE Gender = 'M';
+--------------+
| Average_Age  |
+--------------+
|      25.8333 |
+--------------+
1 row in set (0.003 sec)

mysql>
```

7. Display the name of the Passenger in Confirm status, with maximum age.

himansh1241 — mysql -u root — 92×15

```
mysql> SELECT P_name
    -> FROM Passenger_details
    -> WHERE Status = 'Confirm'
    -> AND Age = (
    ->     SELECT MAX(Age)
    ->     FROM Passenger_details
    ->     WHERE Status = 'Confirm'
    -> );
+----------+
| P_name   |
+----------+
| Abhishek |
+----------+
1 row in set (0.010 sec)
```

8. Display all the students who are female and have confirm tickets and source_id from pari chauk.

himansh1241 — mysql -u root — 92×15

```
mysql> SELECT P_name
    -> FROM Passenger_details
    -> WHERE Gender = 'F'
    -> AND Status = 'Confirm'
    -> AND Bus_no IN (
    ->     SELECT Bus_no
    ->     FROM List_Buses
    ->     WHERE Source = 'Pari Chauk'
    -> );
+--------+
| P_name |
+--------+
| Rupam  |
+--------+
1 row in set (0.006 sec)
```

9. Combine Age and Gender column as of passenger_detail column.

```
himansh1241 — mysql -u root — 92×22

mysql> SELECT
    -> P_name,
    -> CONCAT(Age, '-', Gender) AS Passenger_Detail
    -> FROM Passenger_details;
+----------+------------------+
| P_name   | Passenger_Detail |
+----------+------------------+
| Arjun    | 17-M             |
| Anamika  | 15-F             |
| Divya    | 65-F             |
| Diya     | 19-F             |
| Abhishek | 57-M             |
| Shiva    | 16-M             |
| Rahul    | 21-M             |
| Rupam    | 22-F             |
| Hina     | 23-F             |
| Alam     | 21-M             |
| Satya    | 23-M             |
+----------+------------------+
11 rows in set (0.003 sec)

mysql>
```

10. Write a SQL statement to increase the fare of buses from Pari Chauk and ISBT. The fare will be increased by 50 and 75 for Pari Chauk and ISBT respectively and the fare from for rest of the sources will remain same.

```
himansh1241 — mysql -u root — 92×25

mysql> UPDATE List_Buses
    -> SET Fare = CASE
    ->     WHEN Source = 'Pari Chauk' THEN Fare + 50
    ->     WHEN Source = 'ISBT' THEN Fare + 75
    ->     ELSE Fare
    -> END;
Query OK, 7 rows affected (0.014 sec)
Rows matched: 8  Changed: 7  Warnings: 0

mysql> SELECT *FROM List_Buses;
+--------+-------------+------------+-------------+------+------------+----------+
| Bus_no | Bus_name    | Source     | Destination | Fare | Date       | Time     |
+--------+-------------+------------+-------------+------+------------+----------+
| UP301  | Zing bus    | Pari Chauk | Prayagraj   | 1060 | 2022-08-21 | 21:30:00 |
| HR302  | Cargo       | Sector 62  | Chandigarh  | 1020 | 2022-08-23 | 22:44:00 |
| UP101  | Blue world  | Pari Chauk | Lucknow     | 1290 | 2022-09-24 | 08:05:00 |
| HR303  | Maheshwaram | ISBT       | Ambala      | 1085 | 2022-09-25 | 07:22:00 |
| UP505  | Goluxury    | ISBT       | Kanpur      | 2325 | 2022-09-27 | 19:38:00 |
| DL701  | Vaishnavi   | ISBT       | Chandigarh  | 1625 | 2022-09-28 | 23:55:00 |
| DL306  | Shatabdi    | ISBT       | Dehradun    | 1082 | 2022-09-29 | 20:45:00 |
| UP501  | Safar       | Pari Chauk | Varansi     | 1130 | 2022-09-30 | 08:35:00 |
+--------+-------------+------------+-------------+------+------------+----------+
8 rows in set (0.000 sec)

mysql>
```